ELIM AVPRO + Wing Build: Pack for Matt

Generated from the build set on 2026-10-01 (BOM v8). Don't edit this file; it's rebuilt from `build/`.

1. Scope

ELIM Church runs a portable production rig that deploys once or twice a week at different venues, set up and struck by volunteers under time pressure. This build consolidates the video, network and control gear into a ProX AVPRO rolling workstation (two 12U bays, two 4U tops, a work surface and three lids: the flip-up lid holds both monitors, and the two removable lids pop up as tables). Bay A is the active-system side of the rack; Bay B is the storage and camera-charging side. The Behringer Wing rides in a ProX BXWING hydraulic flip case beside it. Every field cable (cameras, LED wall, confidence monitor, internet) and the five-cable Wing umbilical lands on one labeled, color-coded Neutrik D-series panel on the back of the AVPRO. Almost all the gear inside is already owned. The new spend is the cases, power, cooling, mounts, connectors, cabling and two lid monitors. All cabling is stored with the two cases.

The design is locked. We'd like your quote on the cases and freight, your help pinning a few connector parts, and a sanity check from an outside AV pro.

2. Questions for you

1. **Case and freight quote:** AVPRO price, BXWING price, consolidated freight for both (closed AVPRO is 51.5 × 48.2 × 31.6in). BOM placeholders: $500 / $900 / $250
   When: 2026-10-19. Gates the order? Yes: the order gate
2. **One matched powerCON TRUE1 family:** AVPRO chassis outlet, both umbilical cable ends, and the Wing chassis inlet (only if the Wing panel happens). **Plus the etherCON pair:** NE8FDP-B panel jacks with NE8MX-1 carriers (standard etherCON; the NE8MC-1 Sam named is discontinued and the NE8MX-1 is its successor), checked against the Neutrik compatibility table before the BOM calls them firm
   When: 2026-10-19. Gates the order? Yes, for the connector lines
3. **Rack rails and thread:** tapped 10-32, or square-hole for M6 cage nuts? Both are open. Also open: whether the AVPRO comes with rails built in or they're added after purchase. Do you know? If not, we check the rails at delivery
   When: on case delivery (assumed by 2026-11-15). Gates the order? Screw line only. If you don't know, we order the screws after delivery
4. **BXWING inspection: can it take a panel-mount D-series plate?** Likely doable (2026-10-01), not confirmed. Yes → buy the Wing block. No → the connections run out the side of the case + a TRUE1-to-IEC jumper
   When: on case delivery (assumed by 2026-11-15). Gates the order? Wing block only
5. **Side doors:** do the AVPRO side doors open to usable storage, and what are the inside dimensions? The design doesn't use them
   When: on case delivery (assumed by 2026-11-15). Gates the order? No
6. **Camera-cable trial:** Emily runs the SDI cables we already own (100ft) over a real deploy. Checks pull-through, cable and connector reliability, and picture over length. No camera control. **AOC instead of SDI is an open decision:** about $150 for a 50m run on a spool (estimate), not in the BOM. *Open*
   When: not set. Gates the order? No

7. **The "regret ask":** long-lead items, freight consolidation, anything hard to add later
   When: 2026-10-19 (with the quote). Gates the order? No
8. **Lid legs:** do legs come inside the two removable AVPRO lids? If they don't, we add them as a line item
   When: 2026-10-19. Gates the order? No
9. **ATEM retention tray:** open, not costed. Build a foam or plywood tray, or secure the ATEM another way
   When: not set. Gates the order? Not stated (not costed)

3. Panel maps

AVPRO rear I/O panel

Three Neutrik D-series 12-hole plates (3U) on the Bay A rear rails, behind the upper gear block. This is the one face volunteers touch. Field cables land here; short jumpers run to the gear. Nothing plugs into a device directly. *Decided.*

```
PANEL 1  [CAM1][CAM2*][CAM3][CAM4][CAM5][CAM6][CAM7][CAM8*][CONF OUT][ -- ][ -- ][ -- ]
PANEL 2  [DANTE][LED1][LED2][INTERNET IN][PROD][TALKBACK][IEM][SPARE][ -- ][ -- ][ -- ][POWER > WING]
PANEL 3  [ spare / expansion row, 12 holes ]
         * ATEM IN 2 / IN 8 route open (see the signal map)   -- = blanking plate
```

Jack	Connector	Direction / gender	Inside, goes to	Color
CAM 1, 3–7	BNC 75Ω feedthrough	in	SDI jumper → ATEM IN n	blue
CAM 2, CAM 8	BNC 75Ω feedthrough	in	SDI jumper → ATEM IN 2 / IN 8. Whether the Hollyland RX or the UltraStudio route through them is open	blue
CONF OUT	HDMI feedthrough	out	Anker hub HDMI	white
DANTE TRUNK	etherCON	both	satellite switch	green
LED 1, LED 2	etherCON	out	LED transmitter ports 1, 2	purple
INTERNET IN	etherCON	in	AX21 WAN	red
PROD TRUNK	etherCON	both	AX21 LAN 3	yellow
TALKBACK	XLR **female**	in (a mic plugs in)	trunk to the Wing	orange
IEM	XLR **male**	out (feeds a pack)	from the Wing	orange
SPARE	etherCON	both	AX21 LAN 4 (pre-wired)	yellow
POWER > WING	powerCON TRUE1 **outlet**	out	fed from the CPS1215RM (UPS side)	black

- Order and labels follow layout v15 (POWER > WING last; TRUNK names). *Decided.*
- The POWER jack is an **outlet**: power leaves the AVPRO here. Shore power enters through the UPS line cord, not this jack.
- 18 empty holes get blanking plates. Panel 3 is the growth row (camera returns, more trunks).

Wing panel (conditional)

Planned, pending physical inspection of the BXWING. Nobody has confirmed the case can take a panel, but it's likely doable. Sam inspects on case delivery. The parts are a separate BOM block ($100.00) and are not in the first order. *Conditional.*

```
WING PANEL  [CONTROL][DANTE TRUNK][AES50][TALKBACK][IEM][POWER IN][ -- ][ -- ]
```

Jack	Connector	Gender / direction	To
CONTROL	etherCON	both	Wing control port → PROD trunk → AX21
DANTE TRUNK	etherCON	both	Wing case Dante switch uplink
AES50	etherCON	both	stage box (stage side, not on the umbilical)
TALKBACK	XLR **male**	out to the Wing In 8	mirrors the AVPRO's female
IEM	XLR **female**	in from the Wing Out 8	mirrors the AVPRO's male
POWER IN	powerCON TRUE1 **inlet**	in	pigtail to the Wing's IEC inlet

- **Inside the case:** the Wing case Dante switch (owned TL-SG1005P, 5-port, 3 ports used; 4 while the stream-mix laptop stays in use) and the ProTools Mac. Nothing in the design powers either one yet: the powerCON inlet feeds only the Wing. Open. The ProTools Mac has two links: USB-C to the Wing for recording and a Cat patch to the Dante switch.
- Genders mirror the AVPRO panel: talkback female ↔ male, IEM male ↔ female, POWER outlet ↔ inlet.
- **Fallback if the case can't take a panel:** the connections run out the side of the case to the Wing's own jacks. etherCON ends plug into ordinary RJ45 jacks (Wing control, the Dante switch). The XLRs plug into Wing In 8 / Out 8. Power: the umbilical's female TRUE1 end reaches the Wing's IEC inlet through a short TRUE1-to-IEC C13 jumper (BOM fallback line, qty 0 until needed). The fallback changes the route, not the connectors.

4. Port and signal map

Signal map

Video

From	Cable	To	Color
Wired Pocket 4K (converter)	SDI field, 100ft (owned)	Panel CAM 1 → jumper → ATEM IN 1	blue
Hollyland RX (on the flip-up lid)	SDI, route open	ATEM IN 2	blue
DJI (converter)	SDI field, 100ft (owned)	Panel CAM 3 → jumper → ATEM IN 3	blue
Rented cameras	SDI field	Panel CAM 4–7 → ATEM IN 4–7	blue
ProPresenter Mac → UltraStudio Monitor (placement open)	USB-C, then SDI, route open	ATEM IN 8	blue
ATEM AUX	SDI	HyperDeck HD Mini (records the clean feed)	blue
ATEM OUT 8	SDI	Resi encoder	blue

From	Cable	To	Color
ATEM multiview	HDMI	LEFT lid monitor	white
ProPresenter Mac, native HDMI	HDMI	LED wall transmitter (Bay A U9)	white
LED transmitter, 2 × ethernet	Cat, short	Panel LED 1, LED 2 → LED wall	purple
ProPresenter Mac → Anker hub	HDMI	Panel CONF OUT → confidence monitor (stage)	white
ProPresenter Mac, second screen	route open	RIGHT lid monitor	—

ATEM IN 2 and IN 8. IN 2 is always the Hollyland RX and IN 8 is always the UltraStudio. *Decided.* How each one connects is open. With the Hollyland RX on the flip-up lid, its SDI still has to reach IN 2; running it through the CAM 2 jack is one option. All eight camera jacks stay on the panel, so every switcher input is visible.

Audio and Dante

From	Cable	To	Color
Stage box (Midas DL251)	AES50	Wing (via the Wing panel AES50 jack if the panel happens)	orange boot
Wing Dante card	Cat	Wing case Dante switch (owned TL-SG1005P, 5-port)	green
Wing case Dante switch	DANTE trunk	AVPRO panel DANTE TRUNK → Bay A Dante satellite (new 8-port PoE switch)	green
Satellite switch	Cat, PoE	AVIO (desk). The AVIO is PoE-powered (confirmed)	green
AVIO	XLR pigtails	ATEM audio Ch 1–2	orange
Bianca's Mac, USB-C ethernet adapter	Cat	satellite switch (Dante). It plays music and media over Dante	green
Tyler's ProTools Mac (in the Wing case)	Cat	Wing case Dante switch. It's a Dante endpoint as well as the USB-C recorder, and it runs the live and stream mixes (decided). The stream-mix laptop is still in use today; its retirement is happening separately from the AVPRO plan (open).	green
Bianca's talkback mic	XLR	AVPRO panel TALKBACK → trunk → Wing XLR In 8	orange
Wing XLR Out 8	XLR trunk	AVPRO panel IEM → Bianca's IEM pack	orange
Wing	USB-C	Tyler's Mac (in the Wing case) for recording	—

- Wing In 8 = Bianca's talkback, Out 8 = her IEM, per the current rig's port map. *Decided, 2026-09-29.*

Production network and internet

From	Cable	To	Color
Venue drop or Starlink	Cat field	Panel INTERNET IN → AX21 WAN	red
AX21 LAN 1	Cat	Production switch (TL-SG105)	yellow
AX21 LAN 2	Cat	Resi	yellow

From	Cable	To	Color
AX21 LAN 3	PROD trunk	Panel PROD TRUNK → Wing CONTROL	yellow
AX21 LAN 4	Cat	Panel SPARE (etherCON, pre-wired)	yellow
Production switch	Cat	ATEM, HD Mini (auto-trigger), streaming laptop, Bianca's Anker hub	yellow

- **Port count:** that fills every wired production port (AX21 4 of 4, switch 5 of 5). A future device (Stream Deck host, Mac Mini) needs a bigger switch (a bigger production switch is on the roadmap).

5. Power

Power

The rig runs on whatever a venue gives. It draws well under one 15A circuit, so one works. **Two are preferred:** the UPS on one, the charging side on the other. Never tie circuits together. *Decided.*

Venue power drops

Power reaches the rig area on **two 125ft 12/3 SOOW drops** from wall outlets. *Decided.*

Drop	Wall end	Rig end	Feeds
Drop 1	5-15P plug	one 5-15R connector body	the AVPRO UPS line cord (and the Wing, through the AVPRO)
Drop 2	5-15P plug	one 5-15R connector body	the Bay B CPPDU511 cord. The CPPDU511 passes power through an extension to a quad box at the separate charging station

- **Both drops are built the same:** 125ft 12/3 SOOW, a Hubbell HBL5266C plug, a Hubbell HBL5269C connector body. Either one can stand in for the other. The Hubbell part numbers are assumed, not a ruling; an equal 15A industrial plug and connector body can be substituted.
- **Drop 1 needs one receptacle.** The UPS is the only thing plugged into it; everything else hangs off the UPS.
- **Quad box at the charging station** (4 outlets, 2ft 12/3 tail with a 5-15P). The charging station is a separate table, fed from the Bay B CPPDU511 by an extension that ends in the quad box. *Decided, 2026-09-29.* The extension's details are open and not costed. The BarnDoor part is assumed, not a ruling.
- Voltage drop: 12AWG over 125ft at these loads (2–3A) is about 1%. Fine.
- Plug each drop into a different wall circuit when the venue has two. If not, both on one is fine; the total is still well under 15A.

UPS side (battery-backed)

drop 1 → **CyberPower CP1500PFCRM2U** (2U, pure sine, 1000W) → **CyberPower CPS1215RM** (basic 1U strip, no surge, no meter) → Bay A gear, desk gear, lids, and the **POWER > WING** powerCON outlet → short powerCON umbilical → Wing.

- One UPS backs both cases. The BXWING can't carry its own UPS, so this is the only way to back the Wing. For standalone use the Wing goes on a wall outlet. *Decided.*

- **No surge behind the UPS.** Stacked surge behind a UPS isn't good practice, and Bay B's metered CPPDU511 already gives us a meter. That's why the ProX X-PC10ASTRO was passed on. *Decided.*
- **Laptops:** the two desk laptops plug straight into the **CPS1215RM** (battery-backed), so they keep running if venue power drops. There's no separate laptop strip. *Decided (2026-09-30).*
- The UPS LCD meters the combined booth + Wing load.
- **Outlet count.** The CPS1215RM has 12 outlets. The UPS side has 14 loads: ATEM, HD Mini, Resi brick, LED transmitter, the 8-port PoE switch's brick, TL-SG105 brick, AX21 brick, fan, two monitor bricks, the Hollyland RX brick, the POWER > WING feed cord and the two laptop chargers (they replace the one laptop strip). *Decided (2026-10-01).* Six are wall-warts that block a neighbour. The UPS's own outlets (8 on the CP1500PFCRM2U) take the overflow; if bricks still crowd the strip, 1ft outlet-saver pigtails fix it (not in the BOM).

Charging side (not backed)

drop 2 → **Clear Power CPPDU511** (Bay B front, metered, drawer down-lights) → **Tripp Lite RS1215-RA** (Bay B rear, camera chargers), plugged into a CPPDU511 rear outlet so both read on one meter. *Decided.* The CPPDU511 also feeds the extension to the quad box at the charging station. *Decided; the extension is open.* Dropping the RS1215-RA is an open proposal only; it stays in the BOM.

- Label both Bay B strips **"chargers + lights only"**. No high-draw loads. Keep the total under 15A.

Estimated draw, UPS side

Device	Est.
ATEM TV Studio Pro 4K	~35W
HyperDeck HD Mini	~10W
Resi	~20W
LED wall transmitter	~30W
Production switch + AX21	~20W
Dante satellite switch	~5W
UltraStudio + Hollyland RX	~15W
2 × lid monitors	~50W
2 × laptops (0 when charged)	~0–180W
Wing (over the umbilical)	~90–100W
Total	**~275–465W**

- That's 28–47% of the UPS's 1000W; the high end is both laptops charging. **Spare: ~535–725W (about 4.5–6A at 120V).** Enough for Starlink (~50–75W) and other ad-hoc gear on backup.
- Estimates only. The UPS LCD shows the real number; log it at the first bench test.
- Field devices (confidence monitor, LED wall, cameras and converters) are powered where they sit.
- Utility side: camera chargers and drawer lights. Well under 15A.

Umbilical (AVPRO ↔ Wing)

Five short cables, all color-coded and labeled. The cases always sit side by side. *Decided.*

Cable	Length	Ends	Owned?
PROD trunk	15ft	etherCON both ends	buy (BOM trunk cable + carriers)
DANTE trunk	15ft	etherCON both ends	buy (BOM trunk cable + carriers)
TALKBACK	15ft	XLR	owned (Sam)
IEM	15ft	XLR	owned (Sam)
POWER	15ft, 14/3 SJOOW	powerCON TRUE1 male (AVPRO end), female (Wing end)	buy

- The 15ft length leaves room to angle the cases.
- **Why 14/3 for the power link:** Neutrik lists the TRUE1 cable connector (NAC3FX-W) for 6–12mm cable. 12/3 SOOW (the drop cable) is 15.1mm and won't fit; 14/3 SJOOW is 9.4mm, rated 18A. The Wing draws about 1A.
- **Flexibility:** if a venue ever needs the cases apart, add longer trunks and a longer powerCON link then. Nothing on either panel changes.
- powerCON family: one matched TRUE1 family for the outlet, both cable ends, and the Wing inlet. You or the vendor confirm it before we order.

6. Fill hardware

Quantities and prices are in the attached `BOM.xlsx` (tabs: Buy, Owned, Check, Changes). The last column below is the price status (est / firm / quoted / placeholder / decided), not the price. Cases and freight are left off this list; they're question 1.

- **First order:** $4,641.84, of which $1,650.00 is the case and freight placeholder (AVPRO $500, BXWING $900, freight $250). Without cases: $2,991.84.
- **Wing panel block:** $100.00, bought only if the BXWING can take a panel.

First order

Group	Item	Model / SKU	Qty	Price status
AVPRO panel	1U 12-hole D-series plate	My Cable Mart D-series	3	est
AVPRO panel	BNC 75 ohm feedthrough (camera SDI)	Neutrik NBB75DFI	8	est
AVPRO panel	HDMI feedthrough	Neutrik NAHDMI-W-B	1	est
AVPRO panel	etherCON feedthrough	Neutrik NE8FDP-B	6	est
AVPRO panel	XLR female chassis	Neutrik NC3FD-L-B-1	1	est
AVPRO panel	XLR male chassis	Neutrik NC3MD-L-B-1	1	est
AVPRO panel	powerCON TRUE1 chassis outlet	Neutrik NAC3FPX (family to confirm)	1	est
AVPRO panel	Feed cord for the POWER > WING outlet	12/3 cord with 5-15P, wired to the outlet	1	est

Group	Item	Model / SKU	Qty	Price status
AVPRO panel	D-hole blanking plate	Neutrik DBA-BL	18	est
Spares	BNC feedthrough, spare	Neutrik NBB75DFI	2	est
Spares	etherCON feedthrough, spare	Neutrik NE8FDP-B	2	est
Spares	XLR female chassis, spare	Neutrik NC3FD-L-B-1	1	est
Spares	XLR male chassis, spare	Neutrik NC3MD-L-B-1	1	est
Cable ends	etherCON cable carrier (standard etherCON, over an RJ45 plug)	Neutrik NE8MX-1 (to confirm)	11	est
Cable ends	RJ45 plugs, shielded (pack), for the carrier ends	Shielded Cat6A RJ45 plugs, 10-pk	1	est
Cable ends	powerCON TRUE1 cable connector, male	Neutrik NAC3MX-W (to confirm)	2	est
Cable ends	powerCON TRUE1 cable connector, female	Neutrik NAC3FX-W (to confirm)	2	est
Power	UPS, 2U pure sine (backs booth + Wing)	CyberPower CP1500PFCRM2U	1	est
Power	Rack PDU from the UPS (basic, no surge)	CyberPower CPS1215RM	1	firm
Power	Bay B front PDU, metered, drawer lights	Clear Power CPPDU511	1	firm
Power	Bay B rear strip, camera chargers	Tripp Lite RS1215-RA	1	quoted 2026-08-20
Cooling	1U thermostatic exhaust fan	AC Infinity CLOUDPLATE T1	1	est
Cooling	1U perforated blank panel	Steel perforated (Amazon)	3	est
Mounting	1U vented cantilever shelf, 10in deep	AC Infinity 1U vented (B01C9KYUG8)	3	est
Network	Bay A Dante satellite switch, 8-port PoE	PoE model open	1	placeholder: PoE model not priced
Storage	4U rack drawer	StarTech 4UDRAWER	2	est
Cable mgmt	1U cable lacing bar	StarTech 12S-CABLE-LACING-BAR	1	est
Cable mgmt	Hook-and-loop straps	ONE-WRAP + WrapStrap	1	est
Hardware	Rack screws + cage nuts	M6 or 10-32 (open; check the rails)	1	est
Cabling	SDI BNC jumpers, panel to ATEM	Belden 1855A / premade BNC-BNC	8	est
Cabling	Cat6A patch, color-coded (lot)	Monoprice SlimRun Cat6A 10-pk	1	est
Cabling	Trunk cable, shielded Cat6A, 15ft	Shielded Cat6A patch, 15ft	2	est
Cabling	HDMI 2.0 jumper, short	HDMI 2.0, 6-10ft	3	est
Cabling	Short power cord, 5-15P to IEC C13	2-3ft	6	est
Cabling	Umbilical power cable (AVPRO > Wing), 14/3 SJOOW	14/3 SJOOW, cut to length (ft)	15	est

Group	Item	Model / SKU	Qty	Price status
Venue power	Power drop cable, 12/3 SOOW (2 x 125ft)	12/3 SOOW, cut to length (ft)	250	est
Venue power	Edison plug, 5-15P (wall end)	Hubbell HBL5266C	2	est
Venue power	Edison connector body, 5-15R (rig end)	Hubbell HBL5269C	2	est
Venue power	Quad box, 4 outlets, 2ft 12/3 tail with 5-15P	BarnDoor QB-M-2FT metal quad box	1	decided
Monitors	Lid monitor, 24in matte IPS	LG UltraGear 24GS60F-B	2	est
Monitors	Right-angle HDMI + DC adapters (lot)	90-deg HDMI + DC barrel	1	est

Wing panel block (conditional)

Group	Item	Model / SKU	Qty	Price status
Wing panel	Small D-series plate (8 holes)	D-series plate	1	est
Wing panel	etherCON feedthrough	Neutrik NE8FDP-B	3	est
Wing panel	XLR male chassis	Neutrik NC3MD-L-B-1	1	est
Wing panel	XLR female chassis	Neutrik NC3FD-L-B-1	1	est
Wing panel	powerCON TRUE1 chassis inlet	Neutrik NAC3MPX (family to confirm)	1	est
Wing panel	Inlet-to-Wing power pigtail	IEC C13 cord, cut and wired to the inlet	1	est
Wing panel	D-hole blanking plate	Neutrik DBA-BL	2	est
Wing panel	FALLBACK ONLY: powerCON TRUE1 to IEC C13 jumper	TRUE1 male to C13, short	0	est

7. Layout

ELIM AVPRO - Rack Layout

Build set, 2026-10-01. ProX AVPRO workstation (two 12U bays, two 4U tops, three lids) + the Wing in a BXWING flip case. Bay A = active system; Bay B = storage + utility power.

Camera / SDI = Blue · HDMI = White · Dante = Green · Production net = Yellow · Internet = Red · LED data = Purple · Analog XLR + AES50 = Orange · Power = Black · Blank / reserved

01 Front - operator side

Two lid monitors and the work surface above the bays. Nothing plugs into a device directly.

FLIP-UP LID - 2 monitors (300 nit matte, model open) + Hollyland RX on top

- **LEFT lid** — ATEM hardware multiview, day one. LG 24GS60F-B (model open)
- **RIGHT lid** — ProPresenter Mac second screen. LG 24GS60F-B (model open)
- **Hollyland RX** — on top of the flip-up lid when open (mount open); SDI to ATEM IN 2 (route open)

DESK

- **ATEM TV Studio Pro 4K** — over the 3in grommet (retention tray open); I/O drops into Bay A
- **AVIO (Dante)** — XLR pigtails to ATEM Ch 1-2 | PoE from the satellite switch
- **UltraStudio Monitor** — placement open | ProPresenter USB-C > SDI > ATEM IN 8
- **Streaming laptop** — ATEM Software Control + HyperDeck + Resi
- **ProPresenter Mac (Bianca)** — HDMI > LED Tx | Anker hub > prod ETH + CONF OUT | USB-C > Dante
- **Laptop power** — both laptops plug into the battery-backed CPS1215RM

BAY A - active system (front)

4U top	left open (the Hollyland RX mounts on top of the flip-up lid)
U12	**Fan - 1U thermostatic** — AC Infinity CLOUDPLATE T1; quiet until warm
U11	**Resi** — 1U vented shelf
U10	**HD Mini + Dante satellite (new 8-port PoE switch)** — 1U vented shelf (+ switch power brick)
U9	**LED wall transmitter** — rack-mounted
U8	**Production switch (TL-SG105) + AX21** — 1U vented shelf (antennas up/out)
U7	**Cable management** — lacing bar
U6-4	**Vented blanks (reserve)**

BAY B - storage + utility power (front)

4U top	left open
U12	**CPPDU511 front / RS1215-RA rear** — CPPDU511: metered, drawer lights; fed from drop 2
U11-9	**Left open** — room for a bigger switch or Mac Minis
U8-5	**4U drawer (upper)** — storage

	airflow + growth
U3	**CPS1215RM strip (from UPS)** basic 1U, no surge, no meter
U2-1	**UPS - CP1500PFCRM2U (2U)** pure sine; backs booth + Wing

U4-1	**4U drawer (lower)** heavier items here to balance the case

02 Rear - cable side

Field cables land on the rear I/O panel. Volunteers work from this face. Rear notes show what connects at each device.

BAY A - rear

4U top	left open (the Hollyland RX mounts on top of the flip-up lid)
U12	**Fan - 1U thermostatic** exhausts toward the open back
U11	**Resi** SDI in < ATEM OUT 8 | ETH > AX21
U10	**HD Mini + Dante satellite (new 8-port PoE switch)** HD Mini SDI < ATEM AUX | switch: DANTE TRUNK, AVIO (PoE), Bianca Dante
U9	**LED wall transmitter** HDMI < ProPresenter | 2x ETH > LED 1, LED 2
U8	**Production switch (TL-SG105) + AX21** PROD TRUNK, INTERNET IN, SPARE land here
U7	**Cable management** REAR I/O PANEL (3U) behind the upper block
U6-4	**Vented blanks (reserve)**
U3	**CPS1215RM strip (from UPS)** UPS-backed gear + POWER > WING feed
U2-1	**UPS - CP1500PFCRM2U (2U)** line cord to power drop 1; LCD meters ~275-465W

BAY B - rear

4U top	left open
U12	**CPPDU511 front / RS1215-RA rear** RS1215-RA: camera chargers, plugs into CPPDU511. CPPDU511 also feeds the charging-station extension
U11-9	**Left open**
U8-5	**4U drawer (upper)**
U4-1	**4U drawer (lower)**

03 Rear I/O panel (3U, three plates)

Neutrik D-series. Panel 1 video, panel 2 data / audio / power, panel 3 spare. * = ATEM IN 2 / IN 8 route open (see notes).

PANEL 1 - VIDEO

CAM 1	CAM 2 *	CAM 3	CAM 4	CAM 5	CAM 6	CAM 7	CAM 8 *	CONF OUT	blank	blank	blank
BNC	BNC	BNC	BNC	BNC	BNC	BNC	BNC	HDMI	-	-	-

PANEL 2 - DATA / AUDIO / POWER

DANTE TRUNK	LED 1	LED 2	INTERNET IN	PROD TRUNK	TALKBACK	IEM	SPARE	blank	blank	blank	POWER > WING
etherCON	etherCON	etherCON	etherCON	etherCON	XLR-F	XLR-M	etherCON	-	-	-	powerCON

PANEL 3 - SPARE (expansion)

spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row
-	-	-	-	-	-	-	-	-	-	-	-

04 Wing panel (if the case can take one)

Five umbilical jacks to the AVPRO + one AES50 to the stage box. Genders mirror the AVPRO panel. If the BXWING can't take a panel, the connections run out the side of the case.

WING PANEL

CONTROL	DANTE TRUNK	AES50	TALKBACK	IEM	POWER IN	blank	blank
etherCON	etherCON	etherCON	XLR-M	XLR-F	powerCON	-	-

Inside the case: TL-SG1005P Dante switch (3 of 5 ports used; 4 while the stream-mix laptop stays in use) + the ProTools Mac (USB-C to the Wing, plus Dante; runs the live and stream mixes).

05 Notes

1. Colors = the cable color code. Every cable also carries a number and a source > destination label: match the color, match the number.
2. Bay A is the whole active system. Every network jumper stays in Bay A; nothing crosses bays. Bay B is storage and utility power; no signal. All cabling is stored with the two cases.
3. * ATEM IN 2 is always the Hollyland RX and IN 8 is always the UltraStudio. How each connects is open; running the Hollyland RX through the CAM 2 jack is one option. All eight camera jacks stay on the panel.
4. Power: two 125ft 12/3 SOOW drops from wall outlets (5-15P > 5-15R). Drop 1 feeds the UPS, which backs the booth and the Wing (~275-465W of 1000W with the laptops charging, ~535-725W spare). The desk laptops plug into the CPS1215RM. Drop 2 feeds the Bay B CPPDU511 (+ RS1215-RA for camera chargers), which passes power through an extension (details open) to a quad box at the separate charging station. No surge behind the UPS.
5. Cooling: the open back is primary. The fan exhausts toward the back. Reserve U's are vented blanks. Keep the back clear at venues.
6. Umbilical = 5 short cables (15ft; the cases sit side by side): PROD, DANTE, TALKBACK, IEM and POWER (powerCON TRUE1 on 14/3 SJOOW). AES50 is stage-side.
7. Wing panel is planned, pending inspection of the BXWING at delivery (likely doable). Without it, the connections run out the side of the case to the Wing's own jacks.
8. Dimensions: 12U bays 21.65in deep; 4U tops 4.33in; work surface 49.6 x 16.9in. Three lids: the flip-up lid holds both monitors; the two removable lids pop up as tables (whether legs come with them is open). Side compartments are not used in the design (confirm on delivery).
9. Monitors: 2x LG 24GS60F-B, 300 nit matte. It may not be bright enough even under a tent or canopy, and direct sun needs angling; the model is reopened for brightness. Right-angle HDMI/DC adapters for lid clearance.